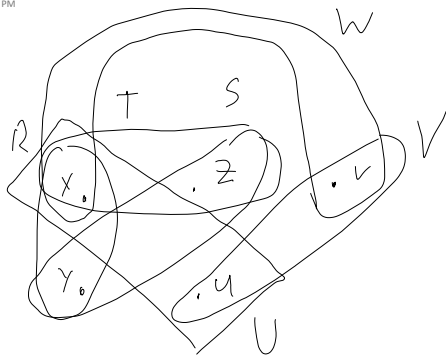


a)



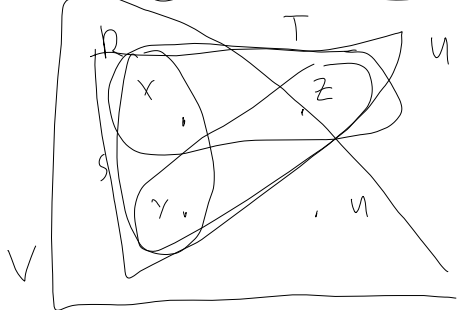
$$1) \begin{array}{c} 2.5 \\ - \end{array} \begin{array}{c} R \quad S \quad T \quad U \quad V \quad W \\ \hline 0 \quad 1 \quad 0 \quad \frac{1}{2} \quad \frac{1}{2} \quad \frac{1}{2} \end{array}$$

$$2) \text{work} = n^{2.5}$$

$$3) \begin{array}{c} TSR (xyz) \\ | \\ RUV (xuv) \end{array}$$

4) $n^{1.5}$ since TSR & RUV are 3-cycles

b)



$$1) \begin{array}{c} 2 \\ - \end{array} \begin{array}{c} R \quad S \quad T \quad U \quad V \\ \hline 0 \quad 0 \quad 0 \quad 1 \quad 1 \end{array}$$

$$2) n^2$$

$$3) \begin{array}{c} ST(x, y, z) - R(x, y) - V(x, y, v) \\ | \\ U(x, y, z) \end{array}$$

4) $n^{1.5}$ STR can be done in $n^{1.5}$, then Yannakakis on V, U , STR in n .